The Hard Problem of Consciousness Assumes Its Conclusion: Why Experience Can Only Be More Structure

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Abstract

The hard problem of consciousness assumes that experience is not structure, then declares that no structural account can explain it. The circularity becomes visible once the demand is stated plainly: explain experience without using structure, when every explanation, concept and bridge must itself be structured. Exposing it needs no experiment: it is an a priori proof in the logician's sense. This paper identifies the assumption, applies the interaction constraint (anything that affects structure must share structure with it) and the rule-random dichotomy (every determinate feature is either ruled or random, and both are mathematical) recursively to every proposed nonstructural alternative. Qualia theories, panpsychism, property dualism and idealism each face their own version of the same problem: specifying what their extra ingredient is, how it combines, and how it interacts with the structure it is supposed to supplement. The structural account needs no supplement. The alternatives need structure to state their own claims. Experience is structure at work: a mind's states producing one another, step by step, as it models its world.

Keywords: hard problem of consciousness, mathematical monism, structural realism, qualia, interaction problem, rule-random dichotomy, panpsychism, property dualism

1. The Demand That Defeats Itself

David Chalmers (1996) asks why physical processing is accompanied by experience. The question presupposes that processing and experience are two different things. If they are, a bridge is needed. Every proposed bridge must specify definite relations between what it connects, and definite relations are structure. The bridge is more math. Chalmers's own bridge shows it: his proposed principles make the structure of experience mirror the structure of awareness (information directly available to guide behavior and report), and give any two systems with the same fine-grained functional organization (the pattern of causal interactions among parts) the same experiences (Chalmers, 1995). Each principle is a relation. What he adds beyond them is that experience is something besides the structure, and that is the separation in dispute.

This paper argues that the hard problem of consciousness, why physical processing should feel like anything, is malformed rather than unsolved. It assumes that experience is nonstructural, then treats the absence of a nonstructural explanation as evidence that something is missing. The missing thing is the assumption, not the explanation.

The argument has three parts. First, the circularity: defining experience as nonstructural disqualifies every structural answer before the investigation starts. Second, the interaction constraint: whatever consciousness is, it must interact with physical structure to cause reports, guide behavior and respond to stimuli, and interaction requires shared structure. Third, the recursive application, repeated at every level, of the rule-random dichotomy: any proposed nonstructural nature, when examined, turns out to be either ruled or random, and both are mathematical. At no stage does a nonstructural remainder appear.

The paper's argument is the eighth link in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside

logic. That is an a priori proof.

2. Every consistent structure exists as mathematics, since anything that let one exist and not another would be structure too, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.

3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.

4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that experience cannot be structure. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The Circular Structure of the Hard Problem of Consciousness

State the hard problem of consciousness as an argument:

First premise: experience has a character that is not identical to any structural property.

Second premise: a structural explanation can supply only structural properties.

Conclusion: no structural explanation can account for the character of experience.

The conclusion follows, but only because the first premise assumes it. That premise is the claim under dispute, not an established fact. Asking whether structure and experience are identical does not require assuming that they are not.

Joseph Levine's explanatory gap (Levine, 1983) may be real, in the sense that we cannot introspectively see why certain organizations feel the way they do. A gap in our view is not a gap in the thing. Temperature and mean molecular kinetic energy seemed like two different things until they were identified. Vitalism posited an extra ingredient for life until biochemistry closed the gap without one. The gap closed when two descriptions turned out to pick out one thing.

A critic who insists that no structural identification could close the consciousness gap has made a stronger claim: that this case is different in kind from temperature and life. That claim needs its own argument. The hard problem of consciousness does not supply one. It restates the assumption.

Those who charge tautology or a lack of evidence fail to understand that this is an a priori proof in the logician's sense: nothing logical can lie outside the logic of

mathematics, and that needs no evidence, any more than 2 + 2 = 4 does. The fundamental truth of reality is likewise not falsifiable by experiment, any more than a mathematical proof is. Like a theorem, it can be refuted only by a counterexample, here one coherent feature that is neither ruled nor random, and the dichotomy leaves no room for one.

Critics sometimes say the structural account defines the problem away. The charge fits the other side. The structural account answers the question it is asked: an organization feels because feeling is what that organization is, running, from the inside, as heat is the motion of molecules, seen at our scale. What it declines is a demand for an answer that uses no structure, which no answer of any kind could meet. The only step that settles the question by definition is the first premise, which simply asserts that experience is something other than structure.

3. The Interaction Constraint

If consciousness is not structure, it faces Elisabeth's question. Princess Elisabeth of Bohemia posed it to René Descartes in 1643: how can an unextended mind (one that occupies no space) move an extended body? His pineal gland, a small gland deep in the brain, answered where the interaction happens, not how. His fallback analogy to weight borrowed a cause he had himself denied was physical.

Put plainly, the question meets anyone who holds that consciousness is an ineffable dimension beyond mathematics and beyond words: how do words reach it? Speech is sound waves. They change what a listener feels, and what the listener feels moves the listener's mouth and sends sound waves back. Physics in, physics out. A consciousness outside the mathematics would need a place where the signal leaves the mathematics and a way for it to translate back.

The question generalizes. For consciousness to cause a report ("I see red"), it must change a physical state. For it to respond to a stimulus, a physical state

must change it. These are causal relations. A causal relation requires definite terms: the thing causing, the thing affected, and the rule governing the connection. Definite terms with governing rules are structure.

A nonstructural consciousness that causes reports is a contradiction: it must have causal structure to cause anything, and causal structure is structure. A nonstructural consciousness that causes nothing is epiphenomenal: it plays no role in producing the very reports that motivated the inquiry. The epiphenomenalist must then explain why a causally inert property tracks the physical processes it was supposed to supplement. The usual answer is a law that pairs each experience with physical features that selection did preserve, so the experience comes along with them. That law says which feature goes with which experience, and a rule that pairs one thing with another is structure.

The interaction constraint is not a separate premise that could be denied. It follows from what interaction means. Two things interact when one makes a determinate difference to the other. Making a determinate difference requires a common system of relations in which differences are defined. That system is structure.

4. The Recursive Argument

The rule-random dichotomy applies to any proposed nonstructural ingredient.

Start with a thing. It has properties; without properties, it is nothing. Each property is some definite way the thing is. Each definite way is either governed by a rule or not. If governed by a rule, the rule is mathematical. If not governed by a rule, the variation is random. Both ruled and random features are mathematical (a rule is a function from states to states, and randomness is defined by probability theory and algorithmic information theory, which measures information by the shortest program that outputs it). The properties are mathematical.

Now examine what each property is. It relates the thing to other things or to possible states. Those relations have their own features. Each feature is either ruled or random. The recursion continues. At no stage does a nonmathematical remainder appear, because at every stage the same dichotomy applies. A proposed substance that is neither ruled nor random is not a third option. It is a refusal to say what the substance is.

Consider the proposed alternatives in turn.

Qualia as primitives. Property dualism adds phenomenal properties, the felt quality of red, to physical properties (Chalmers, 1996). What are these properties? If they are determinate (this red is not that blue), they stand in relations of sameness and difference. Relations of sameness and difference are structure. If they are not determinate, they add nothing specifiable. Calling them "primitive" locates them outside explanation by fiat, precisely the assumption under examination.

If phenomenal properties affect physical events, laws must relate the phenomenal and physical changes. Those laws are structure. If phenomenal properties affect nothing physical, they cannot cause the report "I see red." The physical process produces the report without them.

Panpsychism. Galen Strawson (2006) and Philip Goff (2019) place experience in the intrinsic nature (what a thing is in itself, apart from its relations) that underlies relational structure. What is that intrinsic nature? If it has features, those features are ruled or random, and both are mathematical. If it has no features, it adds nothing to what constitutes the thing. Calling it "experiential" rather than "mathematical" renames structure; it does not escape it.

Panpsychism, the view that experience is a basic feature of all matter, faces a further problem of its own: how do elementary experiential properties combine

into a unified subject? A combination principle must specify which parts combine, under what conditions, and with what resulting experience. Those conditions and results are structure. The combination problem is itself a structural question, and panpsychism has not solved it (Chalmers, 2016).

Russellian monism. Bertrand Russell (1927) observed that physics describes only relational structure, leaving open what instantiates, or embodies, those relations. The Russellian monist fills the gap with phenomenal properties. Yet the filler, if it makes a difference, makes a structural difference. If it makes no difference, nothing has been filled. The question is not whether something sits beneath structure; the question is whether anything coherently can. The rule-random dichotomy says no: whatever the filler is, it is either ruled or random, and both are mathematical.

Idealism. Idealism places mind first and derives matter from it. With one kind of thing, it has no interaction problem between two substances, but it still owes an account of order: an idealist must explain how mental states produce the regularities we call physical law. That explanation requires rules governing which mental states produce which physical regularities. Rules governing transitions are structure. Idealism's "mental substance" turns out to be mathematical structure described from the inside.

5. The Explanatory Asymmetry

The structural account needs no supplement. It must identify which organizations constitute which experiences, and that is research, not a gap in the account. Every alternative needs structure to state its own claims, specify its own interactions and solve its own combination problems. The asymmetry is decisive.

Math needs math. A structural account of consciousness requires structural analysis of organized systems. Descriptions of those systems are themselves

structures (marks on a page, states in a brain, patterns in a computer). All of this is mathematical structure operating on mathematical structure, and no step along the way invokes anything else.

The alternatives need math too. Qualia, the felt qualities of experience, need structural relations to be determinate. Panpsychism needs structural combination principles. Idealism needs structural laws. None of them can state their own position without the very structure they claim to supplement. They cannot explain why everything fits mathematical description, because their own explanations are mathematical descriptions.

The structural account explains why everything fits: because everything is structure. The alternatives cannot explain it; they can only assume an extra ingredient and then describe that ingredient structurally.

6. From Description to Identity

A description of a system and the system itself are different things. Ink on a page does not have the causal powers of the system it describes. This is sometimes offered as evidence that structure is merely representational. It is evidence of the opposite.

The description (ink, pixels, symbols) is one mathematical structure. The system described is another. The description represents the system, and representing is not being: the ink cannot feel what the brain feels, because the ink does not have the brain's organization. The difference is structural, not the presence or absence of a mysterious extra. The brain is a mathematical structure complex enough to model its environment and itself. The ink is not. The test is what follows a change in what the system takes in. Change a running brain's input and the states after it change; the ink stays as written, and a recording plays its stored sequence, even one stored as frame-to-frame differences, whatever happens around it. A

description lists a process; a structure in which each state produces the next one, responding to what it takes in, runs it.

Knowledge by description (modeling a pattern in one cognitive subsystem) and knowledge by acquaintance (instantiating the pattern throughout sensory integration systems) are two different mathematical structures. The first models the second; the second is the experience itself (Jackson, 1982). Mary, who knows all physical facts about color vision through theoretical models but has never seen red, lacks one mathematical structure: the state her brain enters when it processes red light. What she gains on seeing red is a new structural state, not a nonstructural property. She learns no new fact about color; for the first time she is the state her complete theory described, a new way of having what she already knew.

Structure is complete, so two systems cannot differ in experience without differing in structure. The completeness claim is exactly what the hard problem of consciousness denies in advance. Denying it does not refute it. Producing a third option, neither ruled nor random, would.

7. Objections

"You are defining away the problem." The hard problem of consciousness defines away the answer. It stipulates that experience is nonstructural. The present argument examines that stipulation and finds it unsupported. Removing an unjustified assumption is not the same as defining away a problem. The research question remains: which structures constitute which experiences, and why?

"The explanatory gap is real." It may be epistemically real: we cannot see why certain organizations feel the way they do. An epistemic gap, however, does not establish an ontological gap, a gap in reality itself. We could not see why heat was

the motion of molecules before the kinetic theory of heat. The identification did not require us to see it in advance; it required us to stop assuming it was impossible.

"Conceivability of zombies shows experience is not structural." A physical duplicate lacking experience is conceivable only if complete physical organization does not constitute consciousness. That is the premise under dispute. Conceivability arguments test our ability to imagine scenarios; they do not establish metaphysical possibility (what could really be the case) without an independent argument that our concepts track modal space (the range of real possibilities) accurately here, as Katalin Balog argues (Balog, 1999). We can conceive of water not being H2O, but it is.

"The recursive argument proves too much: it would make everything mathematical, including things that are not." That is the claim. Everything is mathematical. The rule-random dichotomy is exhaustive. The objector must produce a coherent third option: something that is neither ruled nor random. No such option has been produced.

"You have not explained why structure feels like anything." Correct. The identity thesis does not explain experience in terms of something else. It says experience is a real aspect of certain mathematical organizations, not something produced by them or added to them. The research question is which organizations and why those. Heat is not explained in terms of something other than the motion of molecules; it is that motion, measured on a different scale. The identity removes the demand for a bridge. The work that remains is finding which systems run those organizations.

8. Conclusion

The hard problem of consciousness is a fallacy of circular reasoning. It assumes that experience is not structure, observes that structural explanations supply only

structure, and concludes that something is missing. What is missing is the justification for the assumption.

Consciousness must be structure to interact with structure. The interaction problem applies to every proposed nonstructural supplement: qualia, panpsychism, idealism and substance dualism (mind as a separate, nonphysical thing). Each must specify what its extra ingredient is (determinate features, therefore structure), how it combines (combination rules, therefore structure) and how it interacts with the physical world (causal relations, therefore structure). The rule-random dichotomy applies recursively at every level. At no stage does a nonmathematical remainder appear.

The structural account has work to do: identifying which organizations constitute experience, explaining why those organizations and not others, and finding which systems have those organizations through comparative, neural and computational study of their structure. That is a research program, not a gap in reality. The alternatives have more work to do: they must state their own positions without using the structure they claim to supplement. So far, none has.

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